Name: Ria Bansal Student ID: 34267433 Date: 27 April 2026

CA Assignment 1 - Assumptions

Surrogate Key Selection

I chose to add surrogate keys to several relations in my logical model. The main reason for this was that some relations had composite natural keys with more than two attributes, which would make them impractical to use as foreign keys in other tables.

The surrogate keys I added are:

- Prop_id in PROPERTY - the natural key (auth_name, street_id, street_no) has three attributes and is referenced in both BIN and PROPERTY_OWNER, so a surrogate key makes this simpler.
- cw_id in CONTRACT_WASTE, cbs_id in CONTRACT_BIN_SUPPLY, bc_id in BIN_COLLECTION, coll_id in COLLECTION, po_id in PROPERTY_OWNER - all have composite keys with more than two attributes.
- owner_id in OWNER - added as a simple unique identifier since owner details like name could change.

In all cases, the original natural keys have been retained as unique constraints to protect business rules.

Minimum Cardinality Assumptions

- A PROPERTY may initially have no bins assigned (stated in the brief).
- A STREET may have no properties on it initially (stated in the brief).
- An OWNER may own zero properties at the time of being added to the system (assumed).
- A TRUCK may exist in the system before it has made any collections (assumed). Therefore, the minimum cardinality on the TRUCK side of the TRUCK-COLLECTION relationship is zero.